

João Ricardo Silva Matos

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Lisbon, Portugal

EDUCATION

• PhD. Student in Computer Science

September 2026 - August 2031

Instituto Superior Técnico, University of Lisbon

School of Computer Science, Carnegie Mellon University

Lisbon, Portugal; Pittsburgh, USA

- Dual degree program jointly conferred by IST and CMU
- Member of the Automated Reasoning and Software Reliability (ARSR) group at INESC-ID

• MSc. in Computer Science and Engineering

September 2023 - November 2025

Instituto Superior Técnico, University of Lisbon

Lisbon, Portugal

- Double specialization in Algorithms and Cybersecurity
- Worked under the supervision of Prof. Pedro Adão
- Dissertation: "Controlling Path Explosion in x86 Symbolic Execution" (19/20) 
- Final grade: 18/20

• Erasmus+

February - July 2024

Akademia Górniczo-Hutnicza (AGH University of Kraków)

Kraków, Poland

- Exchange semester under the Erasmus+ program
- Completed certification for A1 level in Polish

• BEng. in Informatics Engineering

September 2020 - July 2023

Faculty of Sciences, University of Lisbon

Lisbon, Portugal

- Took two extra courses on programming competitions
- Final grade: 17/20

OTHER ACADEMIC ACTIVITIES

• BIP Unite short course in Cybersecurity

October - November 2025

Université Grenoble Alpes

Grenoble, France

- Participated with peers on a case study about cache template attacks
- Worked under the supervision of Prof. Daniel Gruß
- Grade: A

• BIP Unite short course in Artificial Intelligence

September - October 2024

Université Grenoble Alpes

Grenoble, France

- Worked on a case study on sustainable AI
- Grade: 19/20

TEACHING EXPERIENCE

• Instituto Superior Técnico

Various semesters, Sept. 2024 - Now

Teaching Assistant

Lisbon, Portugal

- Forensic Cybersecurity - MSc (2026/27; 2025/26; 2024/25):
 - * Taught weekly lab classes
 - * Prepared and graded the course project
- Software Security - MSc (2026/27; 2025/26):
 - * Taught two lab classes per week
 - * Graded the course project and respective written test
- Introduction to Computer Architecture - BSc (2026/27):
 - * Will teach lab classes in the second semester

EXPERIENCE

• ThinkOpen Solutions [🌐]

Software Developer

September 2025 - Now
Oeiras, Portugal/Remote

- Currently working as a software developer for the Odoo platform
- The developer team creates custom modules according to customer needs
- Projects involve Python, XML, and JavaScript
- Also perform system administration duties

• Premium Minds [🌐]

Software Engineering Intern

July - August 2023
Lisbon, Portugal

- Developed an internal application for employee timekeeping and salary calculations
- Application was created using Java and NextJS

PROJECTS

• Externalization of concretization and state merging policies in symbolic execution

Aug. 2024 - Nov. 2025

Tool: AVD symbolic executor

- Implemented a modified version of the CS_{ML} language for concretization policies
- Created a new language based on CS_{ML} for the specification of merge policies, MC_{ML}
- Modified AVD, a symbolic engine created at Técnico, adding support for these languages
- Created a VSCode extension that supports linting and syntax highlighting for these languages [🐞][🌐]
- Created a testing framework for symbolic execution engines

HONORS AND AWARDS

• Instituto Superior Técnico Academic Excellence Award

2026

Instituto Superior Técnico, University of Lisbon

[🌐]

- Awarded for my academic performance in the 2024/25 academic year
- Award is given every academic year to the top 10% of students (ECTS grade A) with a yearly average of at least 18/20

• Instituto Superior Técnico Academic Merit Award

2025

Instituto Superior Técnico, University of Lisbon

[🌐]

- Awarded for my academic performance in the 2023/24 academic year
- Award is given every academic year to the top 25% of students (ECTS grade B) with a yearly average of at least 17/20

• Faculty of Sciences Merit Award

2022; 23

Faculty of Sciences, University of Lisbon

[🌐]

- Awarded twice for my academic performance in the 2020/21 and 2021/22 academic years
- Award is given to students with a year average higher than 16/20 (1st year) or 16.5/20 (2nd year)

OTHER ACTIVITIES

• Participated in the Guinness World Records' world's largest programming class

2024

Instituto Superior Técnico, University of Lisbon

• Member of the Ribeiro Santos student dorm board

2022-2023

University of Lisbon

• Competed in the MIUP'21 national university programming competition

2021

Instituto Superior Técnico, University of Lisbon

ADDITIONAL INFORMATION

Languages: Portuguese (native), English (fluent, [certified C1](#)), Spanish (fluent), Polish (basic, certified A1)

Interests: Computability; Computational Complexity; Cybersecurity; Formal Verification

REFERENCES

1. **Pedro Adão**

Associate Professor, Department of Computer Science and Engineering
Instituto Superior Técnico
Email: pedro.adao@tecnico.ulisboa.pt
MSc. & PhD. Thesis Advisor

2. **José Fragoso**

Assistant Professor, Department of Computer Science and Engineering
Instituto Superior Técnico
Email: jose.fragoso@tecnico.ulisboa.pt
MSc. Thesis Reviewer & PhD. Co-Advisor

3. **Ruben Martins**

Assistant Research Professor, School of Computer Science
Carnegie Mellon University
Email: rubenm@andrew.cmu.edu
PhD. Co-Advisor

4. **Nuno Santos**

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Forensic Cybersecurity Lecturer